CE 310 — Week 11 Assignment: Hypothesis Testing

Due: Wed Nov 11, 9:00 AM — submit to D2L

Overview

The in-class exercise established that competition moves the price a public agency pays. This assignment pushes on that in two directions: whether the exercise's findings hold on line items it never touched, and whether the competition effect is really about competition — or about where the work happens.

Problem 1 is common to everyone. Problem 2 splits by major. Problem 3 is a short written reflection.

Datasets: `CE310_BidItems_2024_2026.csv` and `CE310_BidProjects_2024_2026.csv` — the same two files as the exercise.

How to Submit

Requirement	Detail
Run all cells	Runtime → **Run all** — wait until every cell shows a checkmark, no spinners
File format	Download as `.ipynb` — File → Download → Download .ipynb (not PDF, not .py)
Major declaration	Set `MAJOR = "CE"` or `MAJOR = "ArcE"` in the Setup cell — this determines which Problem 2 track is graded
Print lines	Do **not** edit or delete any `print(f"ANSWER_... = ...")` line
Written responses	Put each in its own markdown cell immediately after the code cell it responds to
Name and NetID	Fill in `NAME` and `NETID` in the identity cell and run it — that cell is what matches your work to your student record. A blank `NETID` takes a 5-point deduction.
File name	`CE310_W11_Assignment_<NetID>.ipynb` e.g. `CE310_W11_Assignment_abc123.ipynb`

Requirement	Detail
Submission	Upload the `.ipynb` file to D2L → **Assignments** by **9:00 AM, Wed Nov 11**

Problem 1 — Testing Extensions (30 pts)

Do not repeat the exercise computations. The exercise worked one line item, roadway excavation. These questions use items and groupings it never touched.

P1a (8 pts) — Does the Low-Bidder Result Hold on a Different Item?

In A2 you tested whether winning bidders price roadway excavation lower and could not reject H_0. One item proves nothing. Repeat the test on **`MTL W-BEAM GD FEN (TIM POST)`** — metal W-beam guard fence, priced per linear foot.

Work on the **$\log_{10}$ scale**, as the exercise did in B3, and report:

- the t-statistic and p-value
- Cohen's d
- your conclusion at $\alpha = 0.05$

```
ANSWER_P1a_t = ___    # t-statistic, 2 decimals
ANSWER_P1a_d = ___    # Cohen's d, 3 decimals
```

Written (4 pts). In 3–4 sentences: does guard fence behave like excavation or differently? A colleague concludes from your two results that "who wins a bid has nothing to do with the price they quote for any given item." State whether the evidence supports that, and name one thing these tests cannot rule out — remembering what B2 found about the jobs winners bid on.

P1b (8 pts) — The Estimate Benchmark on a New Item

A3 tested whether excavation bids track the engineer's estimate. Run the same one-sample test on **`RC PIPE (CL III)(24 IN)`** — 24-inch reinforced concrete pipe.

Report the mean ratio, the t-statistic against 1.0, and the share of bids strictly above 1.0.

```
ANSWER_P1b_t    = ___    # one-sample t-statistic vs 1.0, 2 decimals
ANSWER_P1b_frac = ___    # share of bids above 1.0, 4 decimals
```

Written (4 pts). The pattern from A3 and A6 repeats here. In 3–4 sentences, state both numbers and explain what feature of the distribution lets a mean significantly above 1.0 coexist with a minority of bids actually exceeding the estimate. Then say which of the two an agency should quote when it reports “how our estimates are performing,” and why.

P1c (7 pts, written) — Which Items Are Priced Alike?

Run Tukey’s HSD on the bid-to-estimate ratio across all **four** line items.

```python
from statsmodels.stats.multicomp import pairwise_tukeyhsd
tukey = pairwise_tukeyhsd(items['Ratio'], items['ItemDescription'], alpha=0.05)
print(tukey)
```

In 4–6 sentences, report how many of the six pairs are significant, identify the **one pair that is not**, and give an engineering reason why those two items might be bid with similar aggression relative to the agency’s estimate while the others differ. Then answer this: you ran six comparisons here and 300 in exercise B9. Explain what Tukey is correcting for, and what would go wrong if you simply ran every pair as an independent t-test at $\alpha = 0.05$.

P1d (7 pts) — Competition as a Grouping Variable

Exercise B7 compared thin competition against deep. That threw away everything in between. Group the awarded contracts by bidder count — 2, 3, 4, 5, 6, 7, and 8-or-more — and run a one-way ANOVA on the winning-bid ratio across all seven bands.

```python
projects['Band'] = projects['NumBidders'].clip(upper=8)
```

```
ANSWER_P1d_F = ___    # ANOVA F-statistic, 2 decimals
```

Written (4 pts). Report F and the p-value. Then address a subtlety: ANOVA tells you the seven band means are not all equal, but a procurement director wants to know whether each additional bidder helps, or whether the benefit arrives all at once and then stops. Look at the band means and say which of those two stories the data supports. Name one thing ANOVA alone cannot tell you here.

Problem 2 — Track Application (18 pts)

Choose one track. CE students complete Track A. ArcE students complete Track B. Declare your major in the Setup cell; only the matching track is graded.

Track A — CE: Is It Competition, or Is It Geography? (18 pts)

Rural districts pay more. That is a familiar complaint in public works, and the usual explanation is that remote work costs more to mobilise and staff. But the exercise showed that thin competition raises prices, and rural districts also attract fewer bidders. Those two explanations predict the same headline and demand different remedies — one calls for a mobilisation allowance, the other for changes to how work is packaged and advertised.

Treat these five districts as metro and the rest as non-metro: `Houston`, `Dallas`, `San Antonio`, `Austin`, `Fort Worth`.

(A2-a, 6 pts) Compare the winning-bid ratio between metro and non-metro contracts. Report both group means, the t-statistic, the p-value and Cohen's d.

```
ANSWER_A2a_t = ___    # t-statistic, 2 decimals
```

(A2-b, 6 pts) Now compare the **number of bidders** between the same two groups. Report both means, the t-statistic and Cohen's d.

```
ANSWER_A2b_t = ___    # t-statistic, 2 decimals
```

In 2–3 sentences, state what A2-a and A2-b together suggest, and why that combination is not yet enough to conclude which explanation is right.

(A2-c, 6 pts) Here is the test that separates them. Restrict to contracts with **3 bidders or fewer** — thin competition everywhere — and compare metro against non-metro within that subset only.

```
ANSWER_A2c_t = ___    # t-statistic, 2 decimals
```

In 4–5 sentences: report the t-statistic, the p-value and Cohen's d, and say what happens to the metro/non-metro gap once competition is held thin. Then state which of the two explanations the evidence favours, what that implies the agency should actually do, and one reason this observational comparison still falls short of proving it.

Track B — ArcE: Does Competition Move the Schedule Too? (18 pts)

> **Why this track is yours.** Contract time is administered by whoever holds the owner's interest during construction — and on building projects that is routinely the architect or the architectural engineer,

> through submittal review, schedule updates, and time-extension requests. The data below is highway work because public agencies publish it at scale and private building owners do not, but the question is one you will meet on a school, a hospital or a tenant fit-out: *when the schedule in the contract was set by market conditions rather than by the work, what does that do to the job you have to administer?* The method transfers; only the vocabulary changes.

Price is not the only thing a contract sets. Contract time — the working days the contractor is allowed — drives staging, escalation exposure and the cost of delay. If competition disciplines price, a natural question is whether it disciplines duration the same way.

(B2-a, 6 pts) Compare contract duration between Construction and Maintenance contracts. Work on the **$\log_{10}$ scale**, since durations span more than an order of magnitude. Report both medians in days, the t-statistic and Cohen's d.

```
ANSWER_B2a_t = ___    # t-statistic on log10(WorkingDays), 2 decimals
```

(B2-b, 6 pts) Something in the maintenance durations is not a measurement. Compute the share of **maintenance** contracts whose duration is exactly 365 days.

```
ANSWER_B2b_frac365 = ___    # share, 4 decimals
```

In 3–4 sentences, explain what that share tells you about how maintenance contracts are written, and what it means for the B2-a comparison — is the Construction/Maintenance duration difference measuring how long work takes, or something else? Say what you would do about it before using this column in a regression.

(B2-c, 6 pts) Restrict to **Construction** contracts only, and test whether duration varies with competition. Run a one-way ANOVA on $\log_{10}$ duration across the seven bidder bands from P1d.

```
ANSWER_B2c_F = ___    # ANOVA F-statistic, 2 decimals
```

Then run a direct two-sample t-test on $\log_{10}$ duration, thin competition (≤ 3 bidders) against deep (≥ 6).

In 4–5 sentences: report both results. The ANOVA and the t-test do not agree at $\alpha = 0.05$ — say which rejects and which does not, and explain how that is possible when both use the same data. Then compare the size of the competition effect on duration against its effect on price from exercise B7, and say what the comparison means for an owner deciding where competition is worth cultivating.

Problem 3 — Written Reflection (12 pts)

Write a **100–150 word** reflection addressing the following prompt:

> Across the exercise and this assignment you have seen the same comparison answered differently depending on how it was framed — a t-test that fails while a rank test succeeds, an ANOVA that fails while a focused two-group test succeeds, and a group difference that disappears once a third variable is held fixed. Describe **one situation** in construction practice where you would trust a hypothesis test to settle a disagreement, and **one situation** where you would insist on seeing the distribution or a subgroup breakdown before accepting the test's answer. Support each with a **specific number** from this week.

Grading criteria (12 pts total): - First situation: specific, plausible, grounded in a number from this week (4 pts) - Second situation: specific, plausible, grounded in a number (4 pts) - Writing quality: concise, within 100–150 words, evidence used appropriately (4 pts)

Grading Summary

Problem	Description	Points
P1a	Low-bidder test on guard fence: t, d, written comparison	8
P1b	Estimate benchmark on RC pipe: t, share above 1.0, written	8
P1c	Tukey across four items; what the correction is for	7
P1d	ANOVA across seven bidder bands; how the benefit accrues	7
P2	Track A (competition vs geography) or Track B (schedule)	18
P3	Reflection, 100–150 words, two situations with numbers	12
Total		**60**

Submission Checklist

- ☐ Setup cell run with `MAJOR` declared
- ☐ P1a: `ANSWER_P1a_t` and `ANSWER_P1a_d` printed + written comparison
- ☐ P1b: `ANSWER_P1b_t` and `ANSWER_P1b_frac` printed + written response
- ☐ P1c: Tukey output shown; non-significant pair identified; correction explained
- ☐ P1d: `ANSWER_P1d_F` printed + written response on how the benefit accrues
- ☐ P2 (only your declared track): all `ANSWER_` values printed and written responses

☐ P3: reflection, 100–150 words, two situations, two specific numbers